AP BIOLOGY TOPIC 1.2

Elements of Life & Atomic Foundations of Biomolecules

UNIT 1: CHEMISTRY OF LIFE (8–11% EXAM WEIGHT)

ENE-1.A | CHNOPS | FUNCTIONAL GROUPS

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§1. KEY TERMS & FORMAL DEFINITIONS LEXICON — TOPIC 1.2 (ELEMENTS OF LIFE & ATOMIC FOUNDATIONS OF BIOMOLECULES)

CED Core Vocabulary

Core Triad (C, H, O) — Most abundant elements in organisms; form the structural framework of carbohydrates, lipids, proteins, and nucleic acids .

Tetravalent — Has 4 valence electrons in its outer shell → forms up to 4 stable covalent bonds (single, double, or triple).

Bonds Readily With — H, O, N, S, and P — high C–C bond dissociation energy (~348 kJ/mol) provides thermodynamic stability + conformational flexibility .

Hydrogen Bonding — When bonded to electronegative O or N, carries a partial positive charge (δ^+) that stabilizes DNA base pairs & protein folding.

Biogeochemical Cycling — Organisms assimilate inorganic forms (CO_2 , H_2O , NH_4^+ / NO_3^- , PO_4^{3-} , SO_4^{2-}) into organic monomers.

Structural Diversity — Forms straight chains, branched chains, and closed rings (isomers with identical formulas but distinct functions).

Hydrogen (H) — Structure + Chemical Energy — Forms 1 single covalent bond (1 valence e^-).

Energy Storage in C–H Bonds — Nonpolar C–H bonds hold high-potential electrons; high H:O ratio → greater free energy yield upon oxidation.

§2. MAJOR BIOELEMENTS (C, H, O, N, P, S)

99% of Biomass

HIERARCHICAL CONCEPT & PATHWAY MAP [CAUSE ➔ MECHANISM ➔ OUTCOME]

Major Bioelements (C, H, O, N, P, S) ➔ **4 classes of biological macromolecules** ➔ **Core Triad (C, H, O)** ➔ **Biogeochemical Cycling**

- Living systems require a constant exchange of matter with the environment to synthesize **4 classes of biological macromolecules**.
- **Core Triad (C, H, O):** Most abundant elements in organisms; form the structural framework of *carbohydrates, lipids, proteins, and nucleic acids*.
- **Biogeochemical Cycling:** Organisms assimilate inorganic forms (**CO_2** , **H_2O** , **NH_4^+ / NO_3^-** , **PO_4^{3-}** , **SO_4^{2-}**) into organic monomers.

§3. CARBON (C): THE VERSATILE BACKBONE OF LIFE

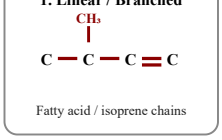
4 Valence e^-

HIERARCHICAL CONCEPT & PATHWAY MAP [CAUSE ➔ MECHANISM ➔ OUTCOME]

Carbon (C) ➔ **Tetravalent** ➔ **Structural Diversity** ➔ **Bonds Readily With**

- **Tetravalent:** Has 4 valence electrons in its outer shell → forms up to **4 stable covalent bonds** (single, double, or triple).
- **Structural Diversity:** Forms straight chains, branched chains, and closed rings (isomers with identical formulas but distinct functions).
- **Bonds Readily With:** H, O, N, S, and P — high C–C bond dissociation energy (~348 kJ/mol) provides **thermodynamic stability + conformational flexibility**.

1. Linear / Branched



Fatty acid / isoprene chains

2. Carbon Ring (Hexose)



Glucose / Steroid rings

3. Key Functional Groups

- **-OH** Hydroxyl (polar)
- **-COOH** Carboxyl (acid)
- **-NH_2** Amino (base)
- **-PO_4^{2-}** Phosphate (neg)
- **-SH** Sulfhydryl (bridge)

[EXAMPLE & BIOLOGICAL APPLICATION] **Glucose rings** (**$C_6H_{12}O_6$**), **fatty acid hydrocarbon chains**, and **steroid 4-ring skeletons** demonstrate carbon's structural versatility.

§6. NITROGEN (N): GENETIC INFORMATION & POLYPEPTIDES

3 Covalent Bonds

HIERARCHICAL CONCEPT & PATHWAY MAP [CAUSE ➔ MECHANISM ➔ OUTCOME]

Nitrogen (N) ➔ **Essential Building Block Of** ➔ **Enables Base Pairing**

- **Essential Building Block Of:**
  - **Amino Acids & Proteins:** Forms the amino group (**-NH_2** / **-NH_3^+**) and links monomers via **peptide bonds** (**-CO–NH-**).
  - **Nitrogenous Bases (DNA, RNA, ATP):** Forms aromatic rings in **Purines (Adenine, Guanine — 2 rings)** and **Pyrimidines (Cytosine, Thymine, Uracil — 1 ring)**.
- **Enables Base Pairing:** Electronegative N atoms participate in specific **hydrogen bonds** between complementary antiparallel strands, enabling faithful genetic replication and transcription.

[EXAMPLE & BIOLOGICAL APPLICATION] **Complementary Base Pairing:**

A–T / A–U form **2 H-bonds**; **G–C** forms **3 H-bonds** (higher G–C % = higher DNA thermal stability / melting temp T_m).

§7. PHOSPHORUS (P): MEMBRANES, NUCLEIC ACIDS & ENERGY

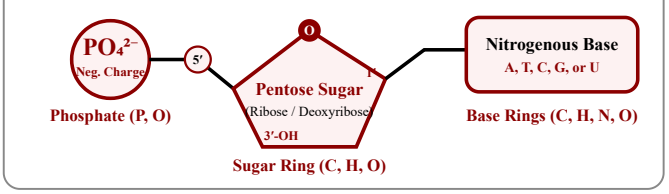
PO_4^{3-} Group

HIERARCHICAL CONCEPT & PATHWAY MAP [CAUSE ➔ MECHANISM ➔ OUTCOME]

Phosphorus (P) ➔ **phosphate group (-PO_4^{2-})** ➔ **In Phospholipids** ➔ **In Nucleic Acids (DNA/RNA)** ➔ **In Cellular Energetics**

- Found biologically as the negatively charged **phosphate group** (**-PO_4^{2-}**), which imparts high water solubility and negative charge at cellular pH (~7.4).
- **In Phospholipids:** Forms the charged, **hydrophilic polar head** that faces aqueous cytosol/extracellular fluid, driving spontaneous **lipid bilayer self-assembly**.
- **In Nucleic Acids (DNA/RNA):** Links the 3′ carbon of one pentose sugar to the 5′ carbon of the next via covalent **phosphodiester bonds**, creating the directional **5′ → 3′ sugar-phosphate backbone** (gives DNA its net negative charge for gel electrophoresis!).
- **In Cellular Energetics:** Forms high-energy phosphoanhydride bonds in **ATP** & regulates enzymes via phosphorylation.

Nucleotide Monomer: Integrating C, H, O, N, and P



Phosphate (P, O) | Sugar Ring (C, H, O) | Base Rings (C, H, N, O)

§4. HYDROGEN (H) & OXYGEN (O): ENERGY & POLARITY

Redox & H-Bonds

- **Hydrogen (H) — Structure + Chemical Energy:**
  - Forms **1 single covalent bond** (1 valence e^-).
  - **Hydrogen Bonding:** When bonded to electronegative O or N, carries a partial positive charge (**δ^+**) that stabilizes DNA base pairs & protein folding.
  - **Energy Storage in C–H Bonds:** Nonpolar C–H bonds hold high-potential electrons; **high H:O ratio → greater free energy yield** upon oxidation.
- **Oxygen (O) — Electronegativity + Energy Transfer:**
  - **Highly Electronegative (3.44):** Pulls shared electrons toward itself (**δ^-**), creating **polar covalent bonds** (hydrophilicity & reactive functional groups).
  - **Cellular Respiration Role:** Acts as the **terminal electron acceptor** in the mitochondrial ETC (**$\frac{1}{2}O_2 + 2e^- + 2H^+ \rightarrow H_2O$**), driving oxidative phosphorylation.

[EXAMPLE & BIOLOGICAL APPLICATION] **Lipids** have long nonpolar C–H chains (low O content) → store **~9 kcal/g** vs. carbohydrates/proteins at **~4 kcal/g**.

§5. MACROMOLECULES & DIAGNOSTIC ELEMENTAL SIGNATURES

High-Yield Radioisotope Table

MACROMOLECULE	ELEMENTS	DIAGNOSTIC RATIO / ISOTOPE MARKER
Carbohydrates	**C, H, O**	**1 : 2 : 1** stoichiometric ratio (**CH_2O**)$_n$; rich in polar **-OH** groups
Lipids	**C, H, O** *(+ P in phospholipids)*	High **C & H**, very little **O** (H:O ≫ 2:1); nonpolar & hydrophobic
Proteins	**C, H, O, N, S**	**N** in every amino group; **^{35}S** uniquely labels proteins (Cysteine/Methionine)
Nucleic Acids	**C, H, O, N, P**	**N** in bases; **^{32}P** labels sugar-phosphate backbone (no S present)

[AP EXAM FRQ SCORING STRATEGY] **Hershey-Chase Experiment**

Application: AP questions frequently ask how to distinguish DNA from protein using radioactive tracers — use **^{32}P for DNA** (phosphodiester backbone) and **^{35}S for Protein** (disulfide-forming amino acids)!

§8. SULFUR (S): TERTIARY PROTEIN STABILITY

Disulfide Bridges

HIERARCHICAL CONCEPT & PATHWAY MAP **[CAUSE ➔ MECHANISM ➔ OUTCOME]**

Sulfur (S) ➔ **Found Exclusively in Proteins** ➔ **Covalent Disulfide Bridges (–S–S–)** ➔ **Functional Role**

- **Found Exclusively in Proteins:** Present in the R-groups of two amino acids — **Cysteine** (contains reactive sulfhydryl **-SH**) and **Methionine** (coded by start codon **AUG**).
- **Covalent Disulfide Bridges (–S–S–):** Oxidation of two **-SH** groups forms a strong covalent cross-link between distant regions of a polypeptide chain.
- **Functional Role:** Locks **3D tertiary (3°) and quaternary (4°) protein conformation** in place, resisting thermal or pH denaturation.

[EXAMPLE & BIOLOGICAL APPLICATION] **Human Insulin** consists of two polypeptide chains (A & B) covalently cross-linked and stabilized by **3 disulfide bridges (–S–S–)**; curly hair keratin is also rich in disulfide bonds.

[CORE AXIOM / REMEMBER]

- **C** → Versatile 4-bond structural **backbone** (chains, branches, rings)
- **H** → Chemical **energy storage** (C–H bonds) & **H-bonding** (**δ^+**)
- **O** → High **electronegativity / polarity** (**δ^-**) & final e^- acceptor
- **N** → **Genetic info** (purines/pyrimidines) & **proteins** (amino groups)
- **P** → **DNA/RNA 5′→3′ backbone**, **ATP**, & amphipathic **membranes**
- **S** → **Protein 3°/4° stability** via Cysteine disulfide bridges (**–S–S–**)

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§9. SCHOLARLY SYNTHESIS: EXPERIMENTAL DESIGN, QUANTITATIVE ANALYSIS, PATHOLOGY & AP FRQ RHETORIC

High-Yield Exam Rigor

MASTER TOPIC SYNTHESIS & CAUSAL ARCHITECTURE — TOPIC 1.2 **[MOLECULAR CAUSE ➔ CELLULAR MECHANISM ➔ PHYSIOLOGICAL / ECOLOGICAL EFFECT ➔ FRQ JUSTIFICATION]**

Topic 1.2: Elements of Life & Atomic Foundations of Biomolecules ➔ **Structural / Energetic Determinants** ➔ **Regulated Pathway & Homeostasis** ➔ **Experimental Perturbation (±2 SE_x̄)** ➔ **Phenotypic / Evolutionary Outcome**

SCHOLARLY DIMENSION	COMPREHENSIVE ACADEMIC REFERENCE & MECHANISTIC SPECIFICATION
I. Experimental Design & Required Lab Controls	AP Lab Context: Macromolecule Indicator Assays (Benedict's for reducing sugars, Lugol's Iodine for starch, Biuret for peptide bonds) & Buffer Titration Curves. Independent Variable (IV): pH or temperature treatment; Dependent Variable (DV): absorbance/colorimetric change or structural stability; Negative Control: deionized H_2O. **Null Hypothesis (H_0):** State explicitly that the independent variable treatment has *no statistically significant effect* on the dependent variable; any observed difference is due to random sampling chance alone.
II. Quantitative Formulas & Graph Interpretation	Quantitative & Graph Analysis: Stoichiometric monomer-to-polymer condensation: for n monomers joined in a linear chain, (n − 1) covalent bonds form and (n − 1) H_2O molecules are released. pH scale is logarithmic: $pH = -\log_{10}[H^+]$; each 1-unit decrease in pH represents a 10-fold increase in $[H^+]$. Caloric density: Lipids (~9 kcal/g) vs. Carbohydrates/Proteins (~4 kcal/g). **Error Bar Rule (±2 SE_x̄):** If the ±2 SE_x̄ error bars of two treatment groups *do not overlap*, there is a statistically significant difference between means ($p < 0.05$; reject H_0).
III. Pathology, Mutation & Disruption Case Study	Clinical & Molecular Disruption Case Study: Missense mutation in the β-globin gene (Glu6Val in Sickle-Cell Disease) replaces a polar/charged glutamic acid R-group with a nonpolar hydrophobic valine on the protein exterior, driving abnormal quaternary hydrophobic aggregation into rigid fibers under deoxygenated conditions. Similarly, extreme fever (>40°C) or acidosis (pH < 7.0) disrupts weak hydrogen and ionic bonds in tertiary (3°) protein structure while leaving primary (1°) peptide bonds intact.
IV. AP FRQ Task Verb Scoring Templates	• **DESCRIBE / IDENTIFY:** State the direct trend, value, or structural component with exact units from the figure. • **EXPLAIN (How/Why):** Connect cause to effect via a 3-link chain: *"[Change in X] causes [molecular/cellular mechanism Y], which results in [observable outcome Z]."* • **PREDICT:** State the specific directional change (increase, decrease, or remain constant) when the pathway is perturbed. • **JUSTIFY:** Pair quantitative data/evidence from the prompt with the underlying biological principle from Topic 1.2 to prove why the claim holds.

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